Python Assignment
Topic-8: Control Statements (Break, Continue, Pass)

Q1. Understanding the role of break statement in Python loops.

Ans.

The break statement is used to stop a loop immediately. When Python finds a break statement, it exits the loop even if the condition is still true. It is useful when we want to stop the loop based on a condition. Break helps in saving time and avoids unnecessary repetitions. It is often used in searching and checking conditions. Using break makes programs more efficient and faster.

Example:

```
for i in range(1,6):
  if i == 3:
    break
  print(i)
```

Q2. Understanding the role of continue statement in Python loops.

Ans.

The continue statement is used to skip the current iteration of a loop. When Python sees continue, it skips the remaining code and moves to the next iteration. It is useful when we want to ignore some values in a loop. Continue helps in controlling the flow of loops. It makes programs clean and easy to understand. It is commonly used when applying conditions inside loops.

Example:

```
for i in range(1,6):
  if i == 3:
    continue
  print(i)
```

Q3. Understanding the role of pass statement in Python.

Ans.

The pass statement is used as a placeholder in Python. It does nothing when executed. It is used when a statement is required but we do not want to write code yet. Pass helps in avoiding errors when writing incomplete code. It is commonly used in empty functions, loops, or conditions. It is useful during development when logic is not ready.

Example:

```
for i in range(3):
    pass
```